Data & AI

11. Inside ResNet18

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Last Time

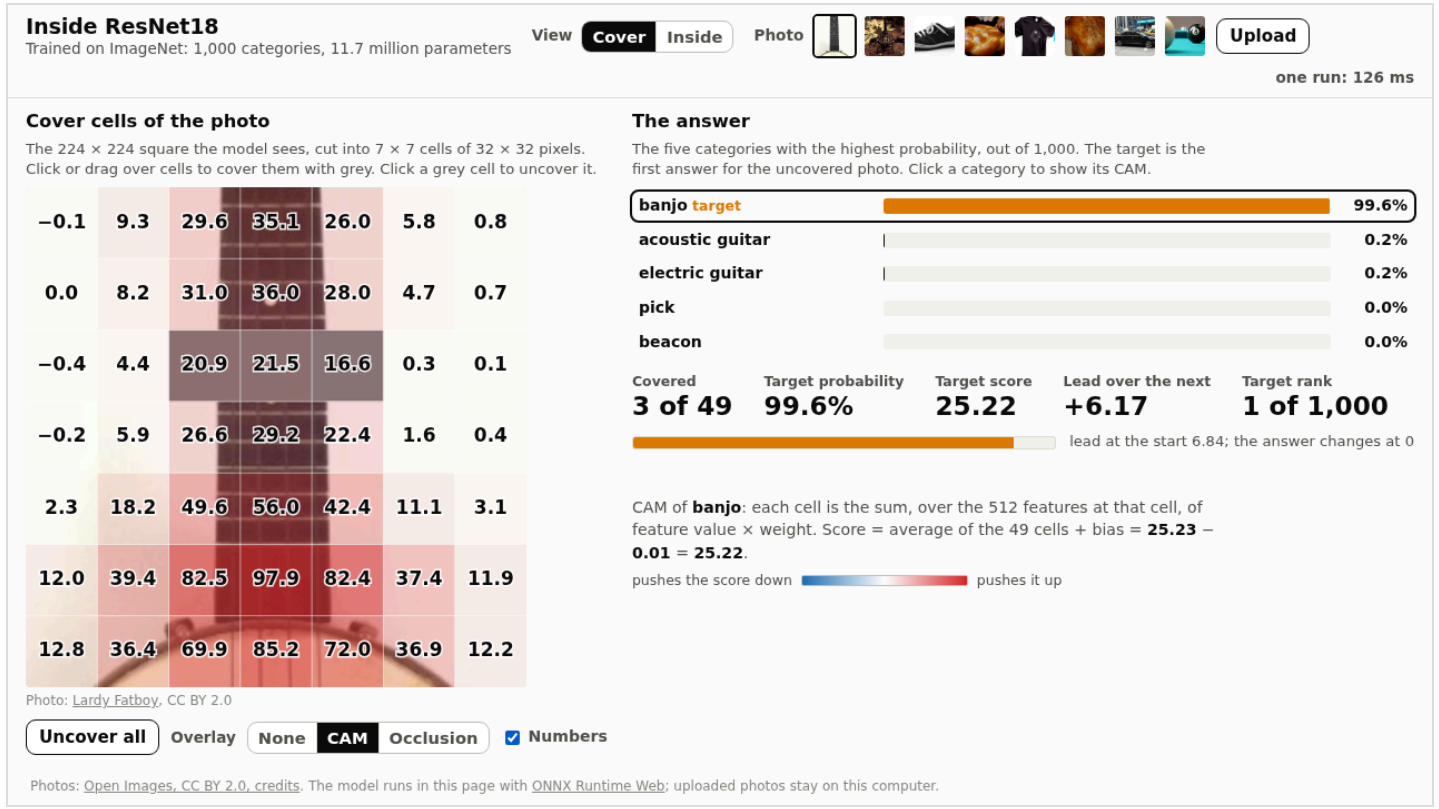
- A small CNN: convolution, max pooling, then 784×10
- ResNet18: 20 convolution layers, 11,689,512 parameters

Today

- Inside ResNet18: what changes its answer, and why
- Gradient ascent on the pixels: feature visualization, DeepDream, style transfer
- Adversarial examples

Inside ResNet18

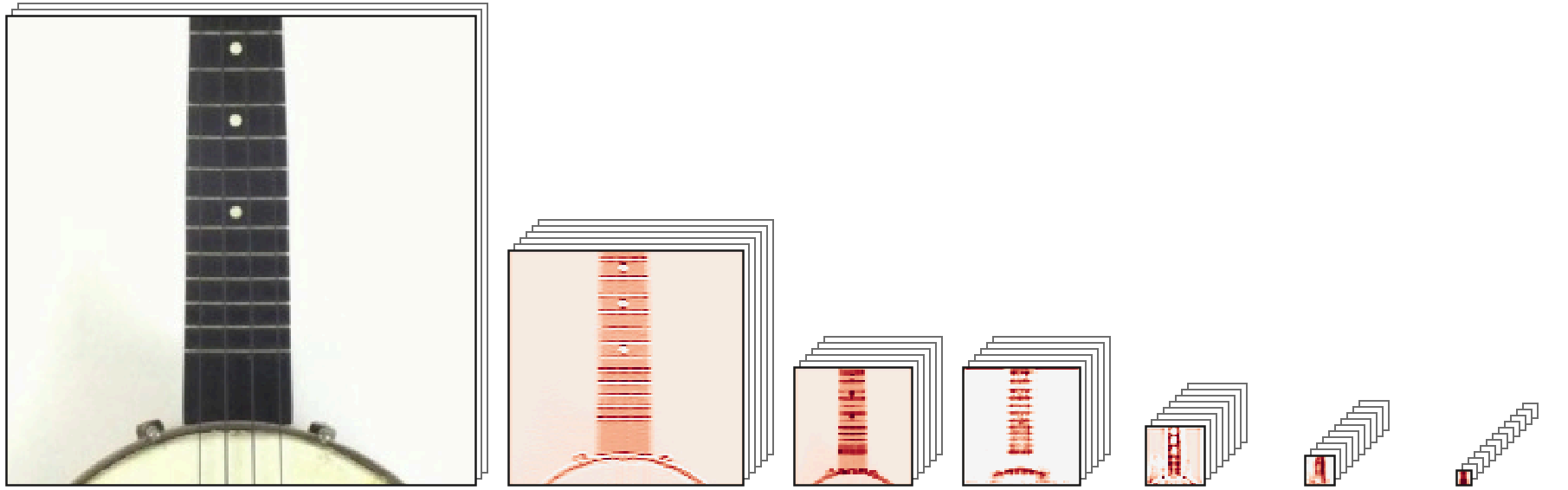
jdasam.github.io/mas1004/demos/lecture11-resnet18-cam



The Structure of ResNet18

ResNet18 Layer by Layer

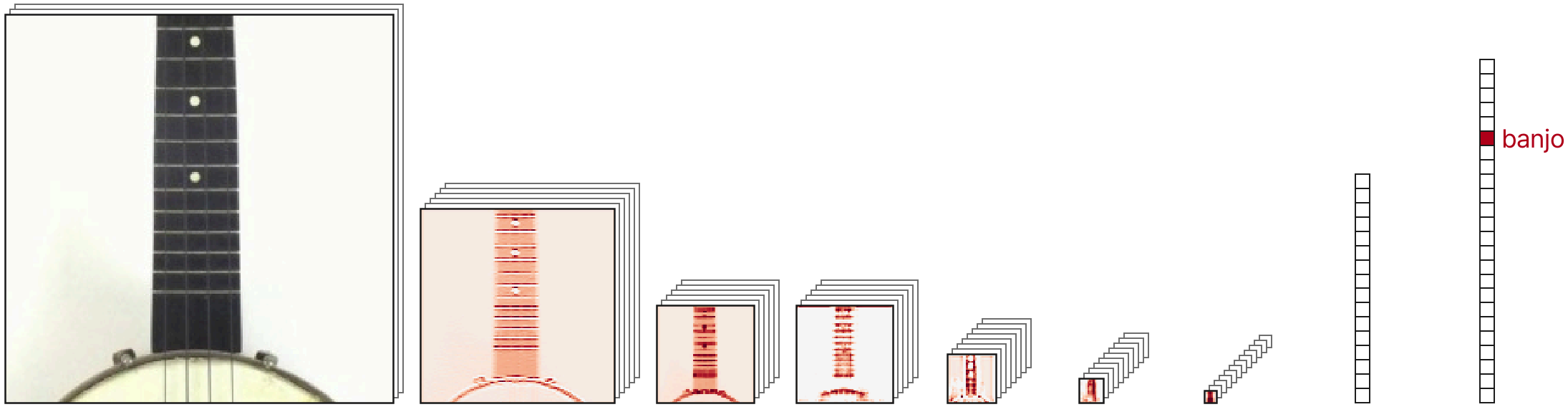
Output shape after each stage, for one 224 × 224 photo



input	conv1	max pool	layer1	layer2	layer3	layer4
224 × 224	112 × 112	56 × 56	56 × 56	28 × 28	14 × 14	7 × 7
3 colours	64 maps	64 maps	64 maps	128 maps	256 maps	512 maps

ResNet18 Layer by Layer

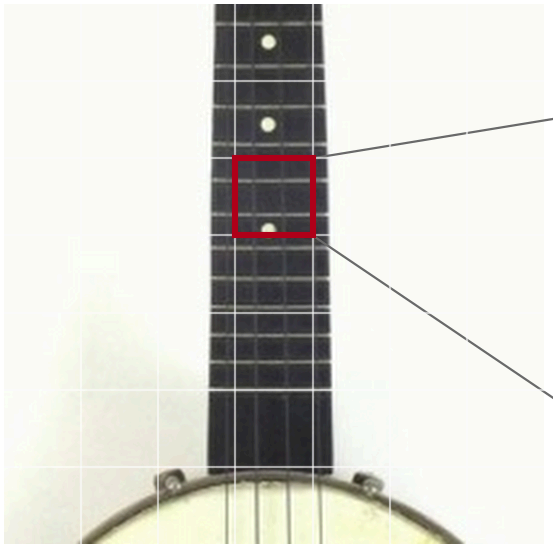
Output shape after each stage, for one 224 × 224 photo



Receptive Field

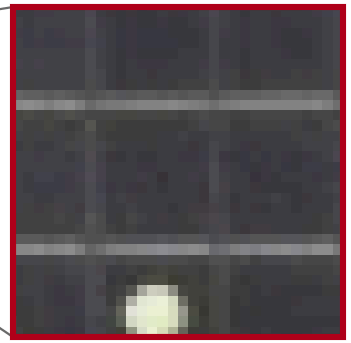
The pixels that can change one cell of layer4

7 × 7 cells



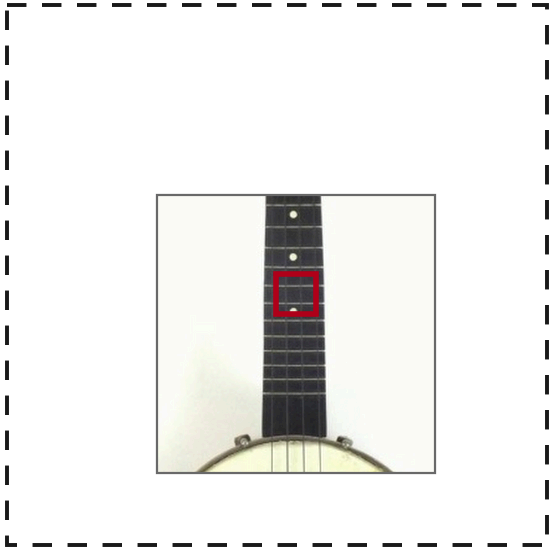
32 × 32 pixels each

row 3, column 4



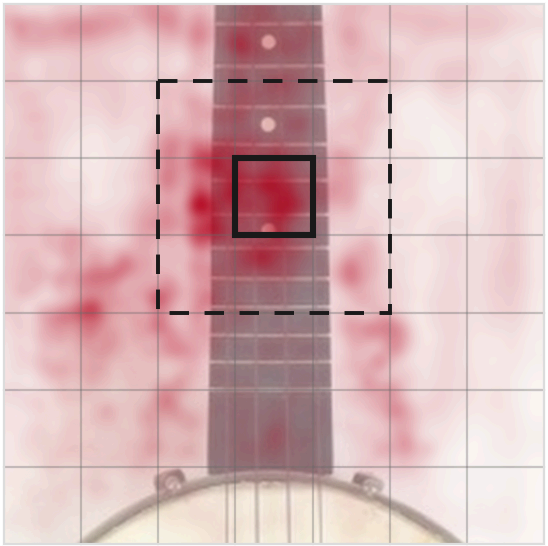
its 32 × 32 pixels

pixels that can change it



435 × 435 pixels,
more than the photo (224 × 224)

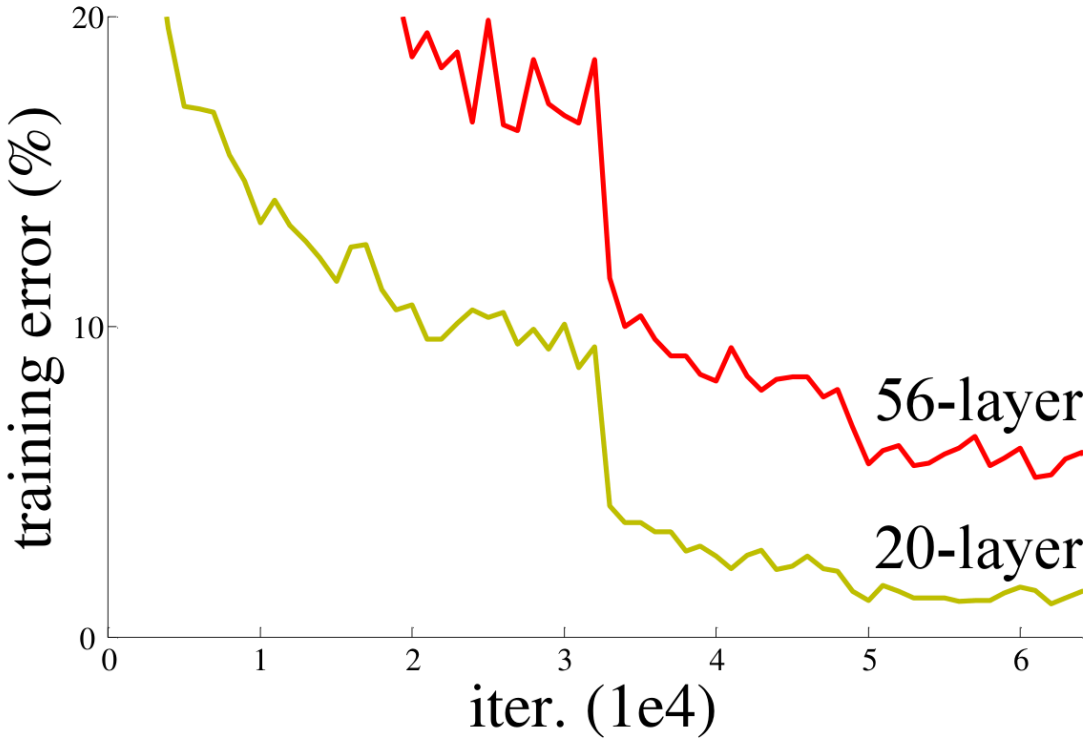
how much each pixel changes it

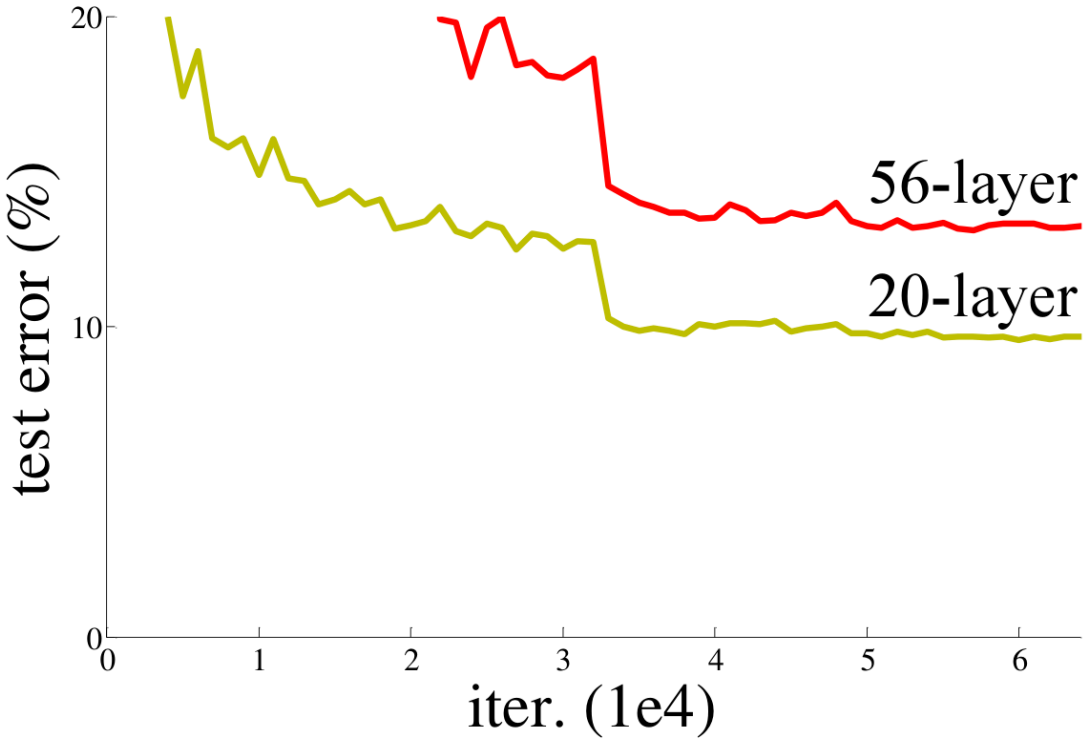


6% inside the cell,
33% inside the 3 × 3 cells

Training Deeper Networks

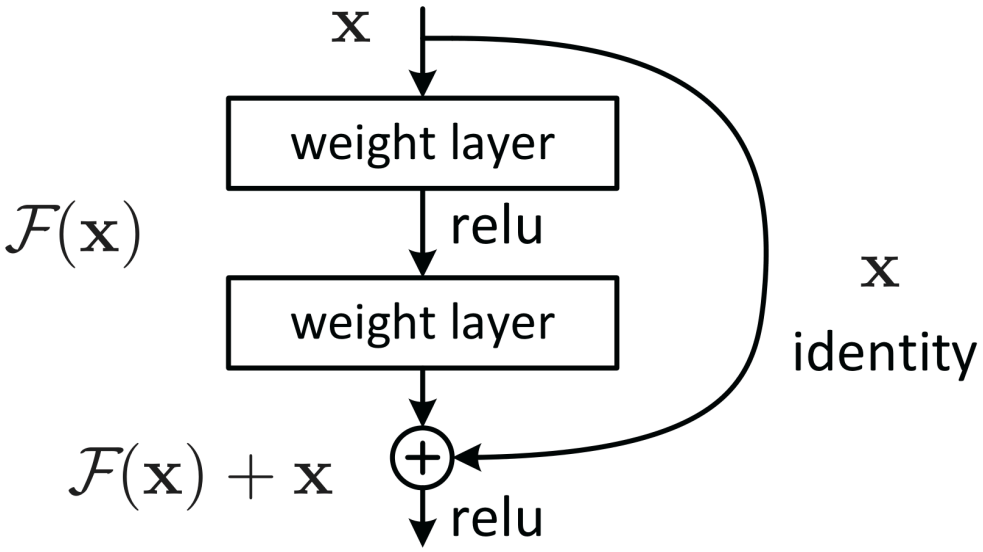
He et al. (2016): plain networks of 20 and 56 layers on CIFAR-10





Residual Block

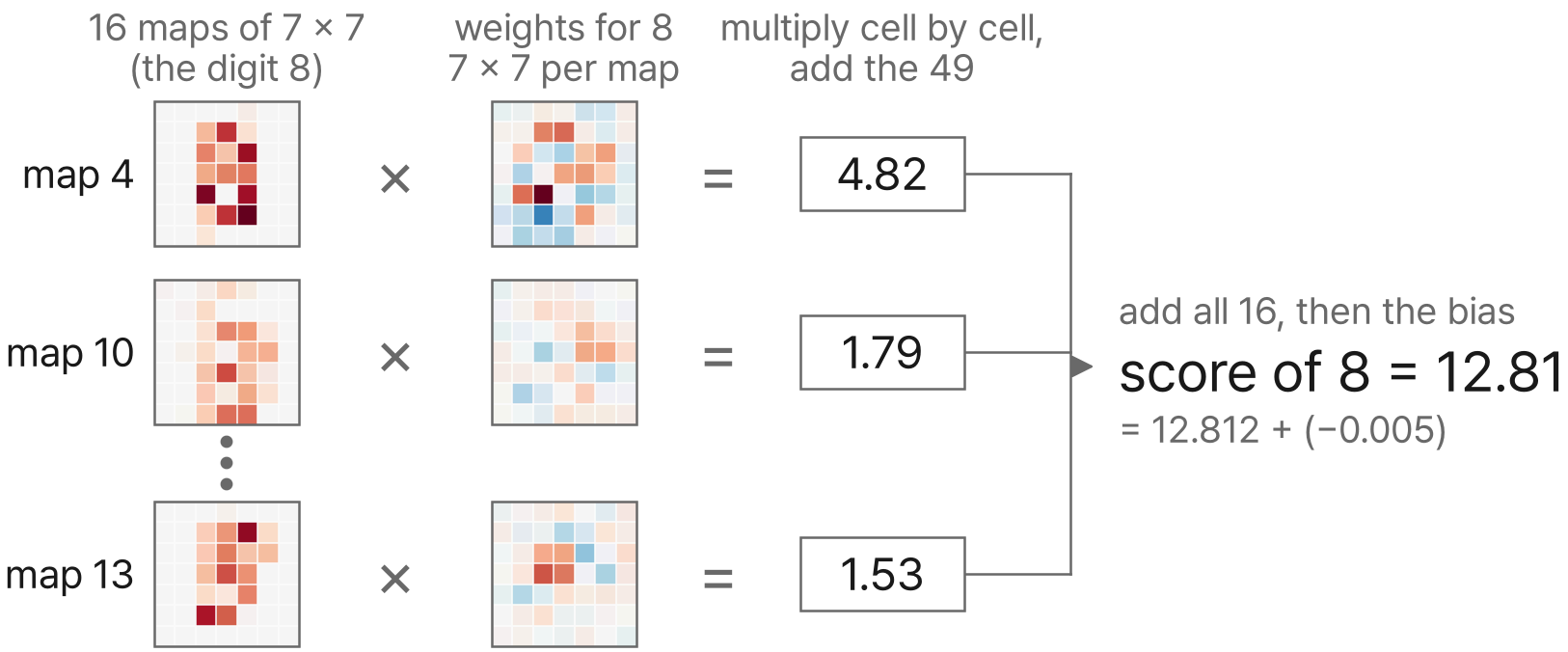
He, Zhang, Ren, Sun (2016), Deep Residual Learning for Image Recognition, CVPR



- ResNet18: 8 blocks of two 3 × 3 convolutions

The Last Layer of the Small CNN for MNIST

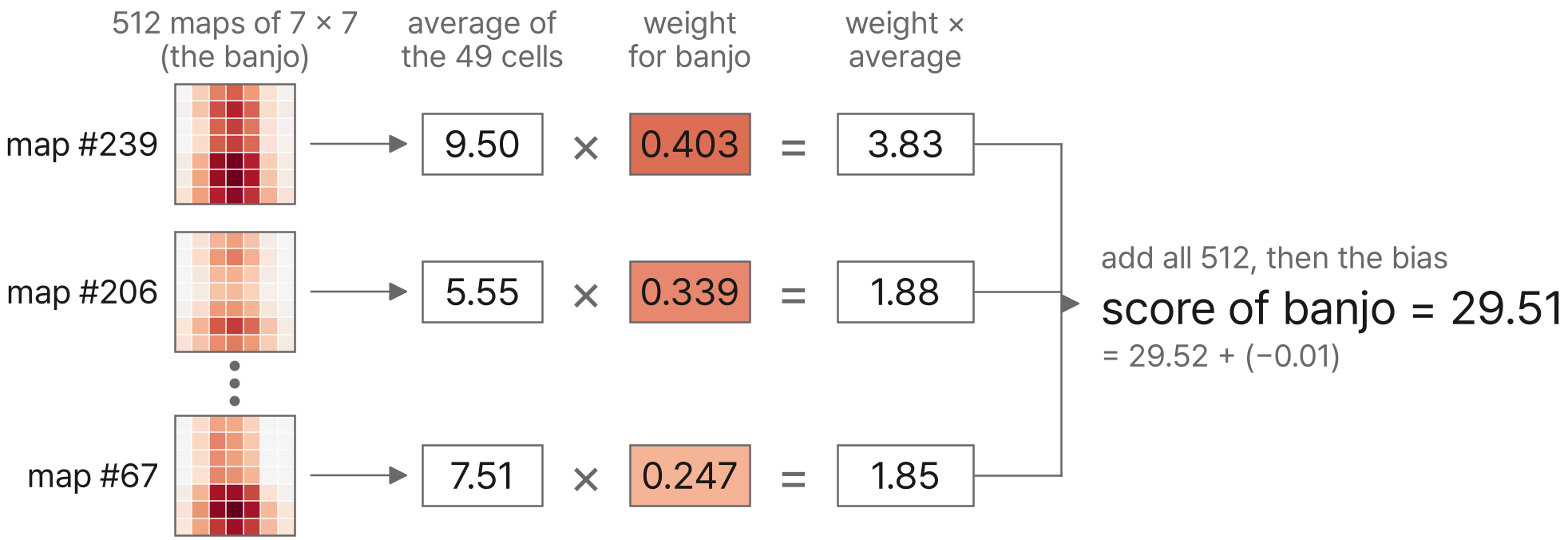
Lecture 9, handwritten digits: 16 maps of 7 × 7 flattened into 784 numbers, then 10 scores



Flattened: 16 × 7 × 7 = 784 numbers, each with its own weight
784 weights × 10 digits = 7,840 weights

Global Average Pooling

ResNet18: each 7 × 7 map averaged into one number, then 512 × 1,000



Averaged: 512 maps become 512 numbers, one weight each (the same for all 49 cells)
512 weights × 1,000 categories = 512,000 weights

Splitting the Score by Cell

$f_k(c)$: value of feature k at cell c · w_k: its weight for the category

$$\text{score} = \sum_{k=1}^{512} w_k \cdot \frac{1}{49} \sum_{c=1}^{49} f_k(c) \; + \; b$$

Splitting the Score by Cell

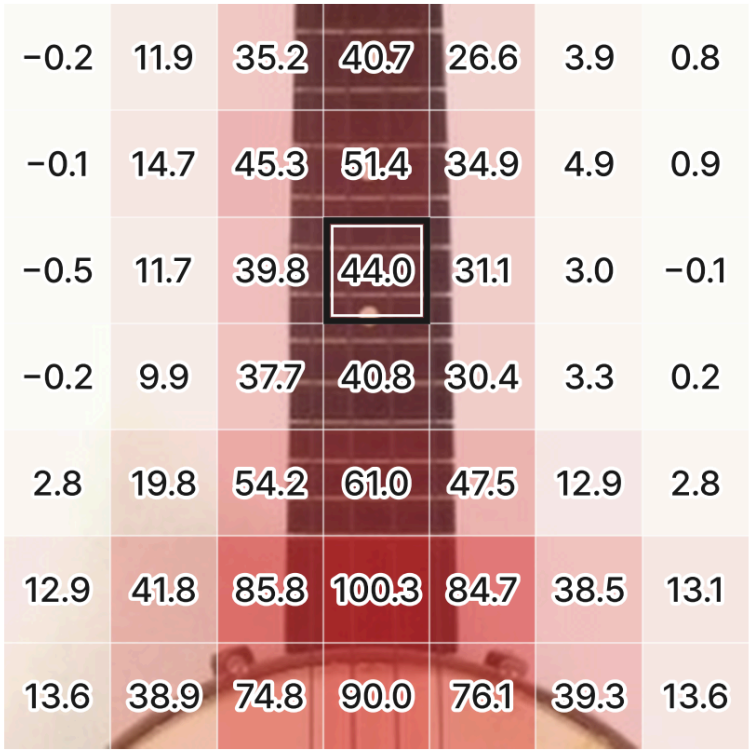
$f_k(c)$: value of feature k at cell c · w_k: its weight for the category

$$\text{score} = \sum_{k=1}^{512} w_k \cdot \frac{1}{49} \sum_{c=1}^{49} f_k(c) \; + \; b$$

$$= \frac{1}{49} \sum_{c=1}^{49} \underbrace{\sum_{k=1}^{512} w_k f_k(c)}_{\text{value of cell } c} \; + \; b$$

Class Activation Map

Zhou, Khosla, Lapedriza, Oliva, Torralba (2016), Learning Deep Features for Discriminative Localization, CVPR



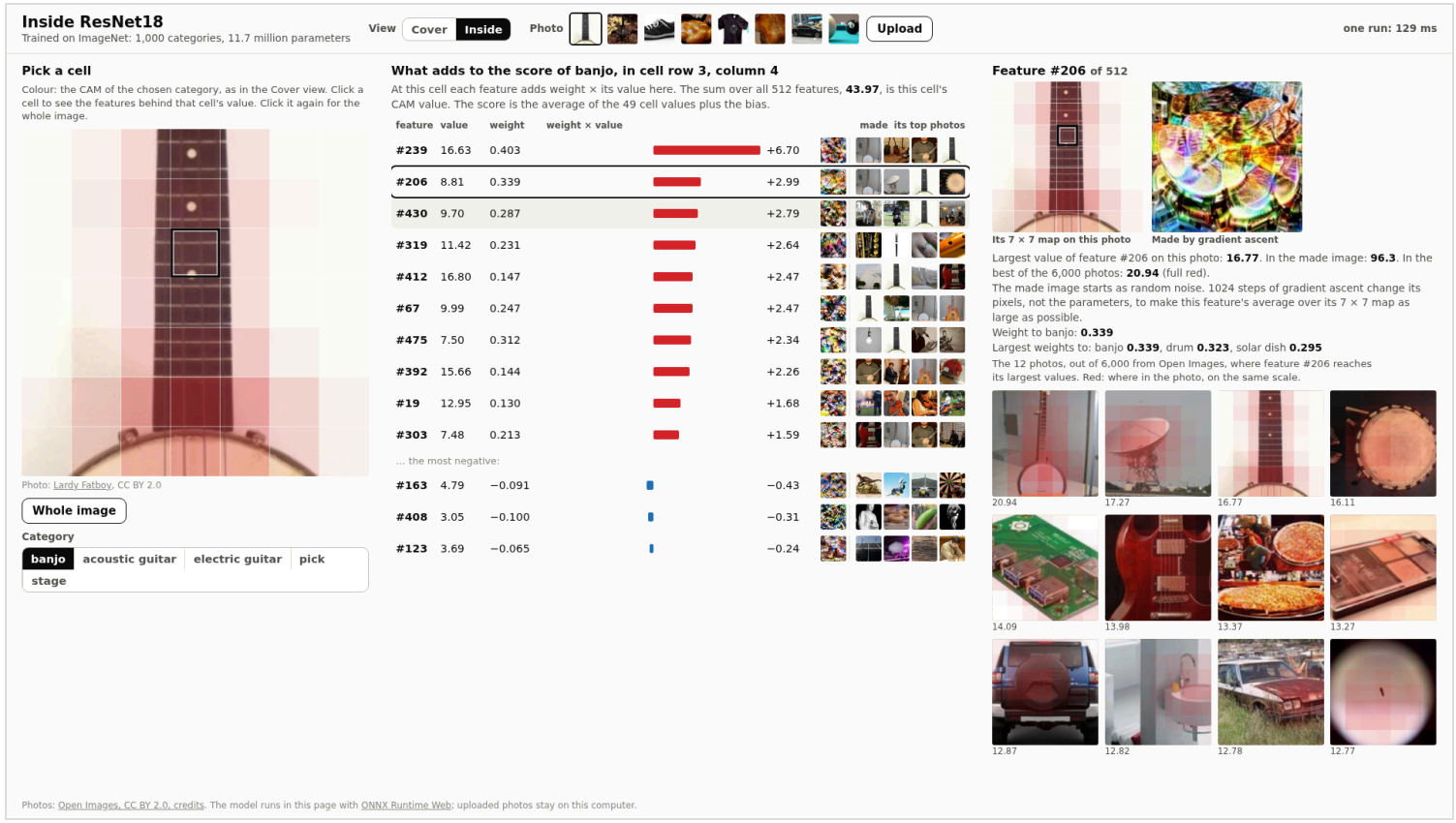
In each cell: weight × value, added over the 512 features

average of the 49 cells	29.52
bias	−0.01
score of banjo	29.51

pushes the score down — pushes it up

Splitting a Cell by Feature

Inside view: banjo, cell at row 3, column 4

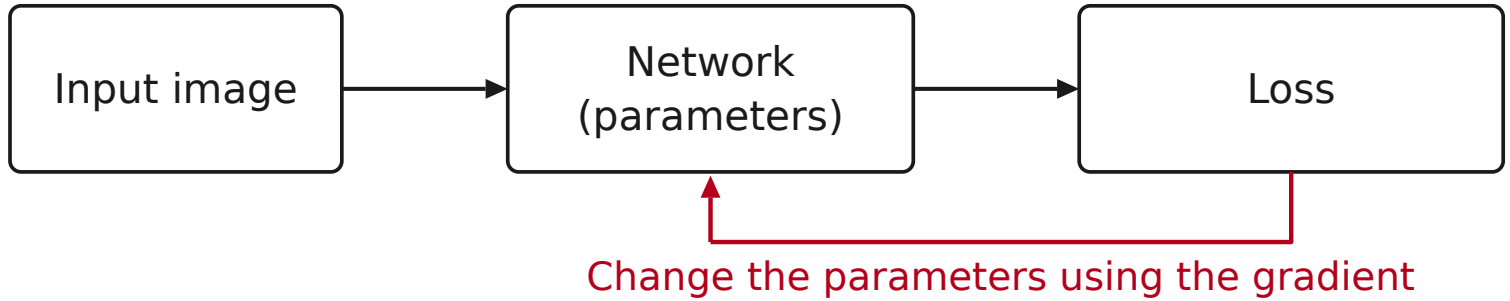


Gradient Ascent on the Pixels

Gradients for the Pixels

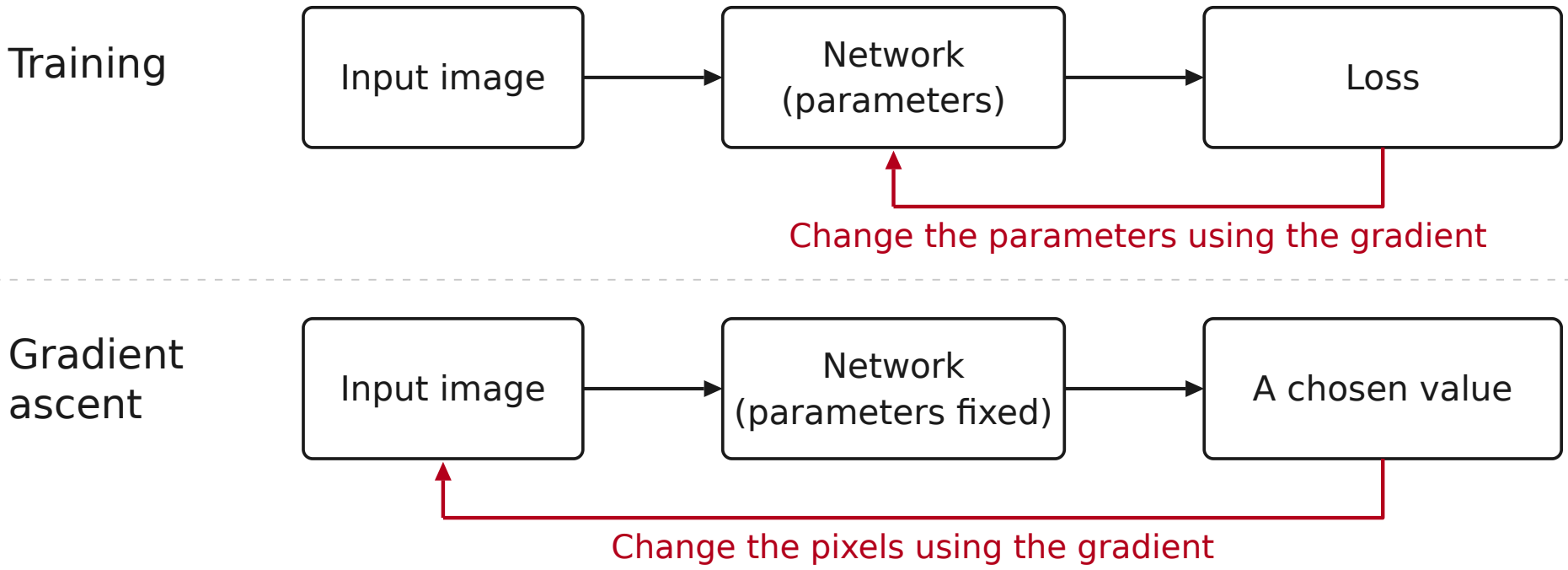
Backprop gives a gradient for every input number too

Training



Gradients for the Pixels

Backprop gives a gradient for every input number too

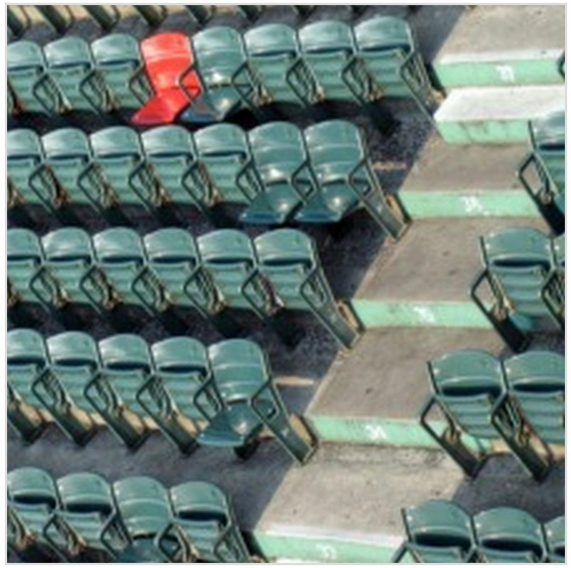


Feature Visualization

Olah, Mordvintsev, Schubert (2017), Feature Visualization, Distill

Best of 6,000 photos

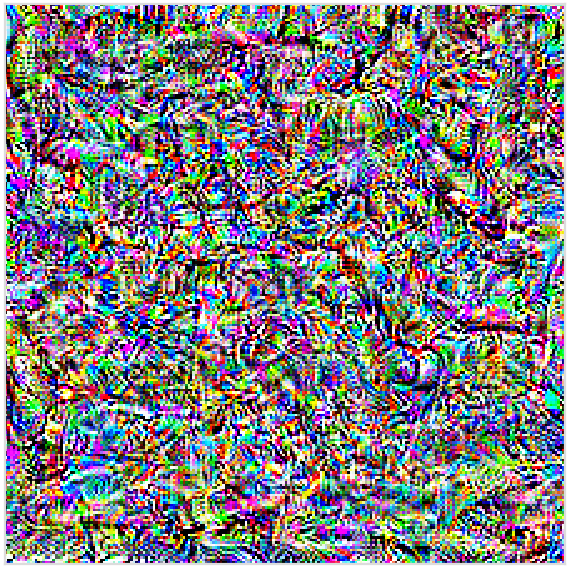
the photo where this feature is largest



feature #194: 10.4

Plain gradient ascent

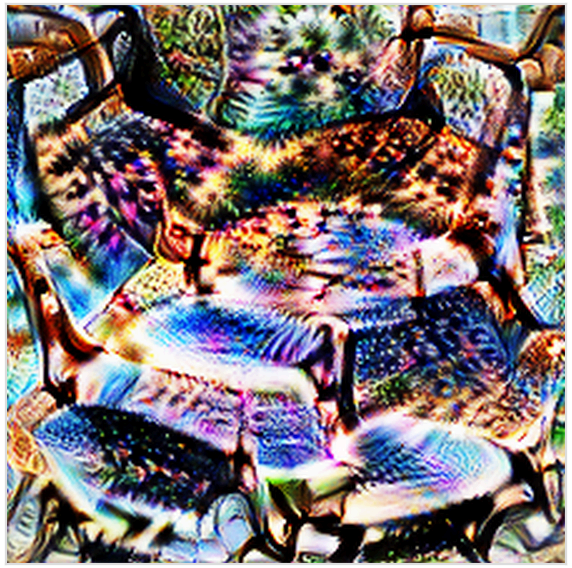
start: random pixels, 1,024 steps



feature #194: 195.9

Regularized gradient ascent

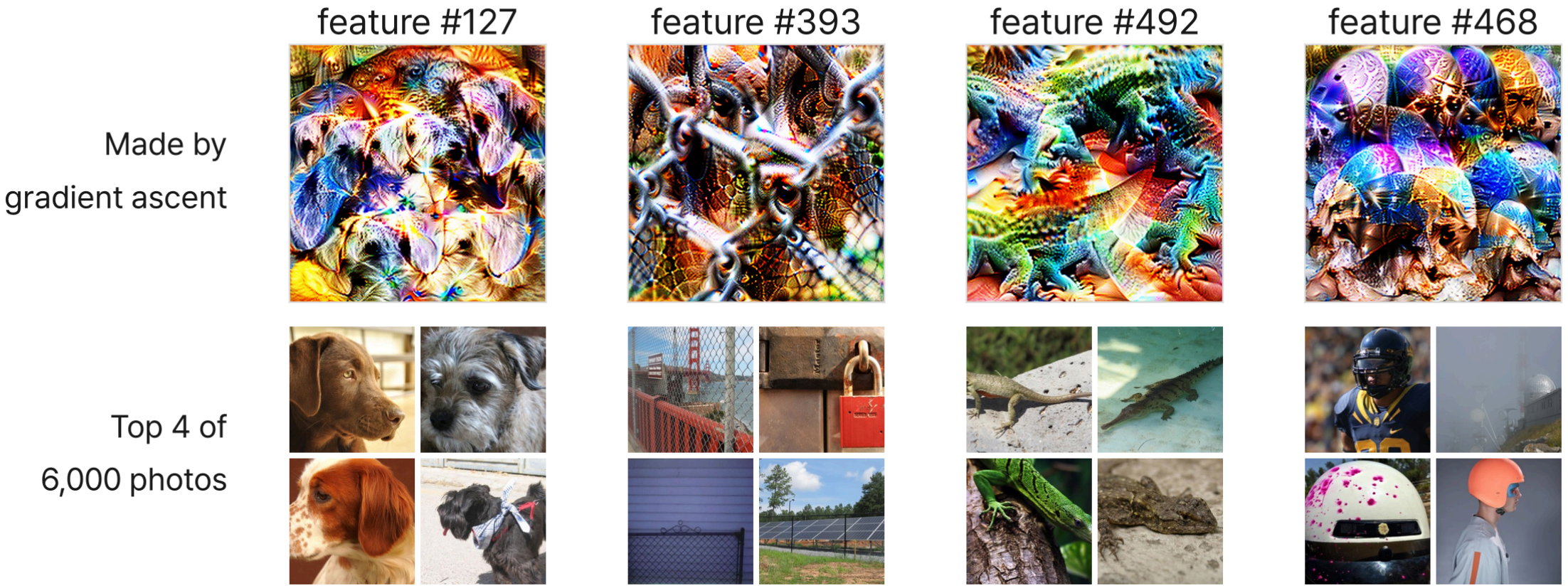
smooth pixels, random shift, scale, rotation



feature #194: 71.0

Features and Their Top Photos

Made by gradient ascent, and the 4 photos out of 6,000 where the feature is largest



Photos: Open Images V7, CC BY 2.0, credits

Class Visualization

Mordvintsev, Olah, Tyka (2015), Inceptionism: Going Deeper into Neural Networks, Google Research Blog



Class Visualization with ResNet18

Gradient ascent on the score of one category

dumbbell



score 142
probability 0.99

banjo



score 200
probability 1.00

pool table



score 142
probability 1.00

cab



score 108
probability 1.00

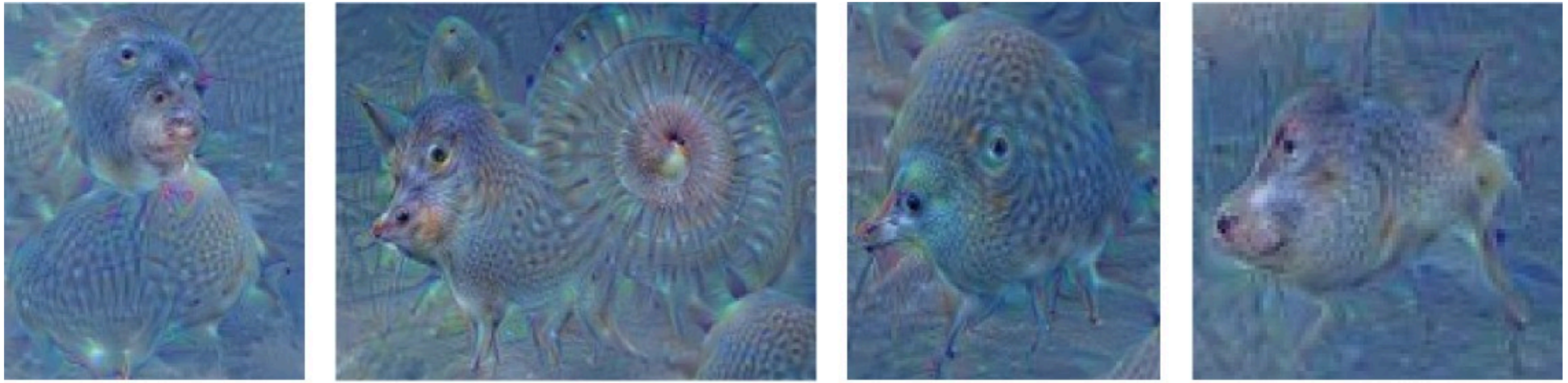
zebra



score 114
probability 1.00

DeepDream

Mordvintsev, Olah, Tyka (2015), Inceptionism: Going Deeper into Neural Networks, Google Research Blog



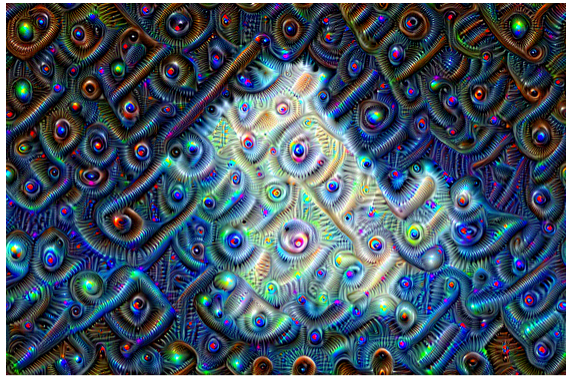
- Start from a photo, and make the values of a whole layer larger

DeepDream by Layer

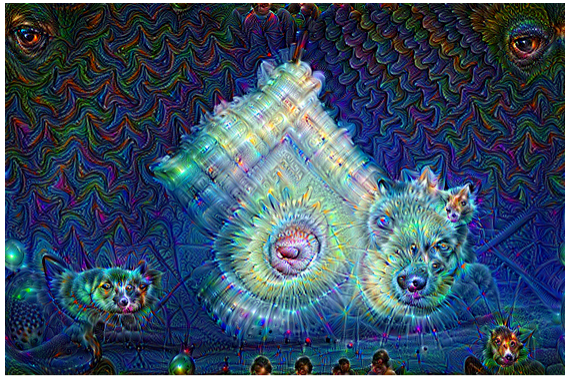
ResNet18, the same photo



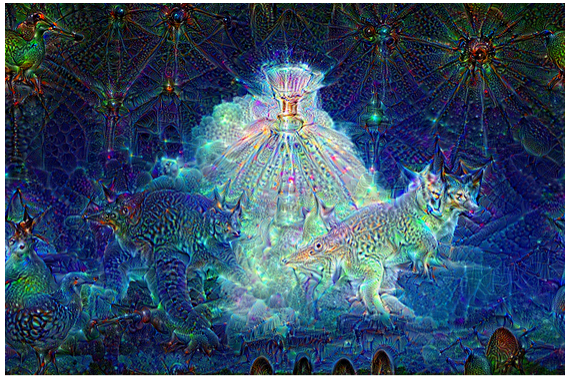
Input photo



layer2 (shallow)



layer3 (middle)

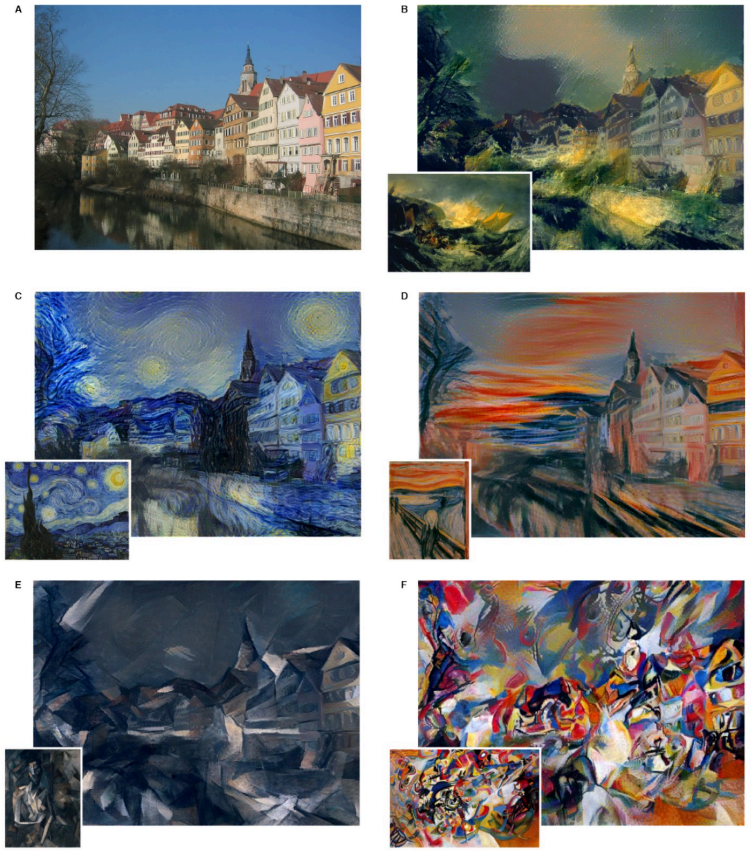


layer4, first block (deep)

Input: Ralph F. Kresge, Cumulus6 (NOAA, public domain)

Neural Style Transfer

Gatys, Ecker, Bethge (2016), Image Style Transfer Using Convolutional Neural Networks, CVPR



- Content loss: the maps of a deep layer, cell by cell
- Style loss: which features appear together, position thrown away
- Change the pixels to lower the sum of the two losses

Style Transfer Result

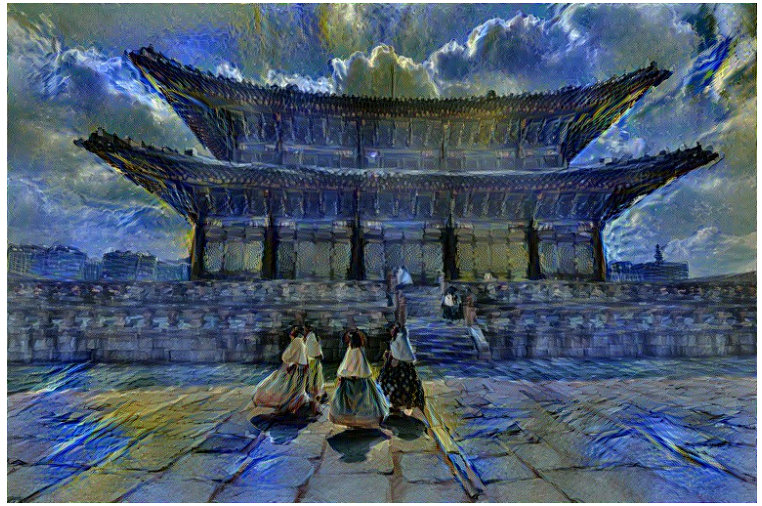
VGG19 · Gyeongbokgung photo and The Starry Night



Content



Style



Result after 400 steps from noise

Content: chsyang, Gyeongbokgung Geunjeongjeon (Public Domain Mark) · Style: Vincent van Gogh, The Starry Night

Adversarial Examples

Adversarial Examples

Goodfellow, Shlens, Szegedy (2015), Explaining and Harnessing Adversarial Examples, ICLR

- The same gradient for the pixels, used to change the answer
- Every pixel moves by a tiny amount that a person cannot see

An Adversarial Banjo

ResNet18, the banjo from the demo, each pixel changed by at most 1 level out of 255



banjo 99.8%



Change × 60

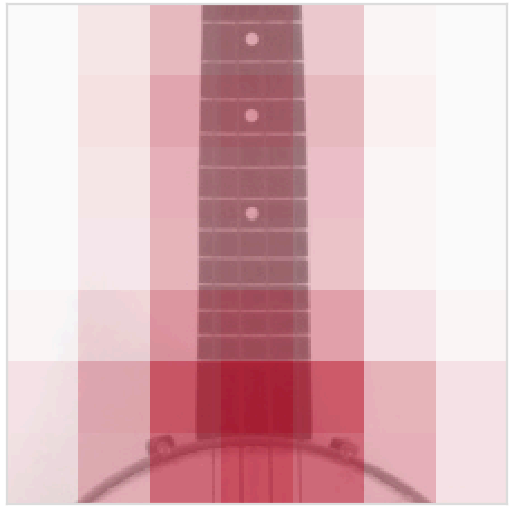


toaster 100%

CAM of the Adversarial Banjo

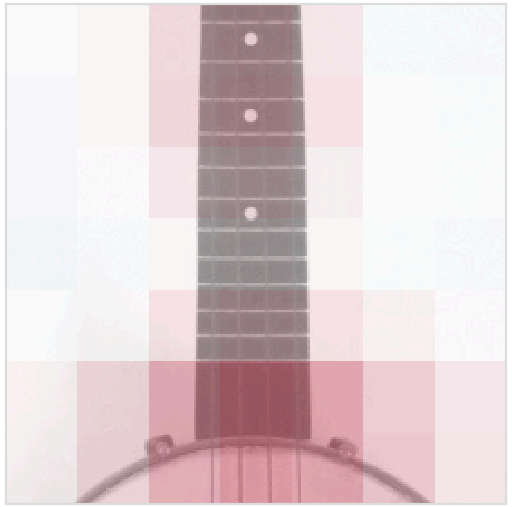
The same model, before and after the change

banjo, original photo



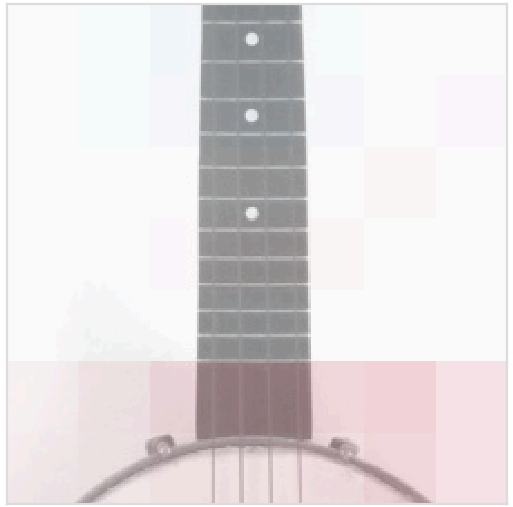
score 29.5

banjo, changed photo



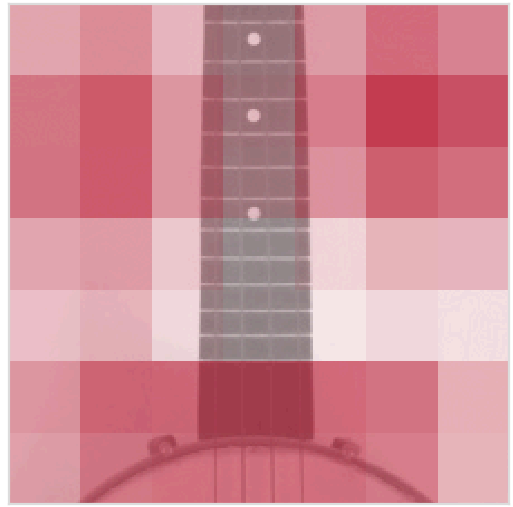
score 11.4

toaster, original photo



score 2.0

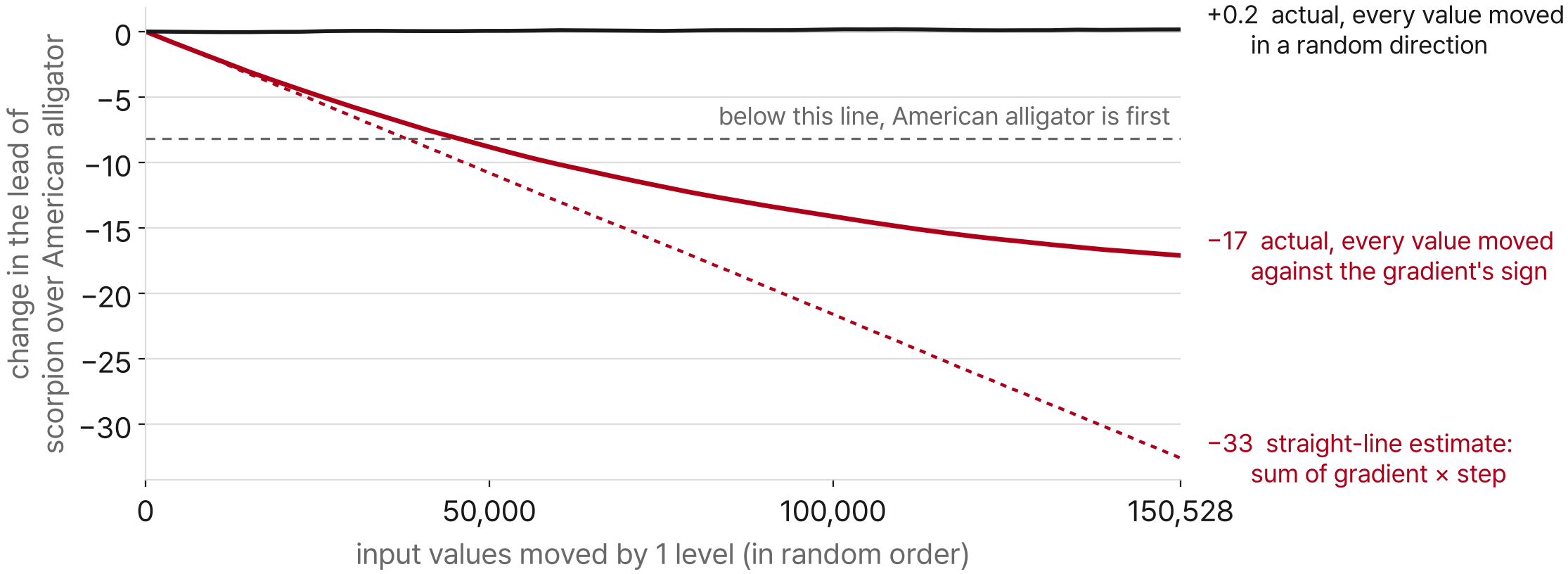
toaster, changed photo



score 31.2

Many Small Changes Add Up

Scorpion photo from the demo: 3 × 224 × 224 = 150,528 input values



Coming Up

- Assignment 1 due Friday, October 16, 22:00